PRESENTAION OF TARNSMISSION LINE

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SUBJECT - POWER SYSTEM - II

UNDER THE GUIDANCE -

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Why We Need Transmission Line





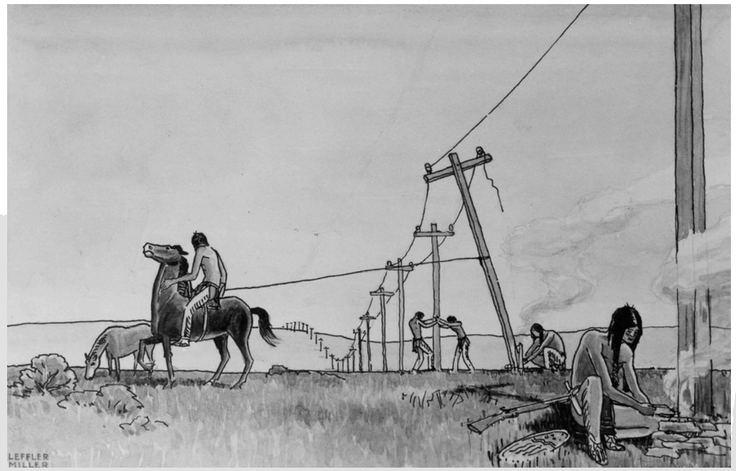


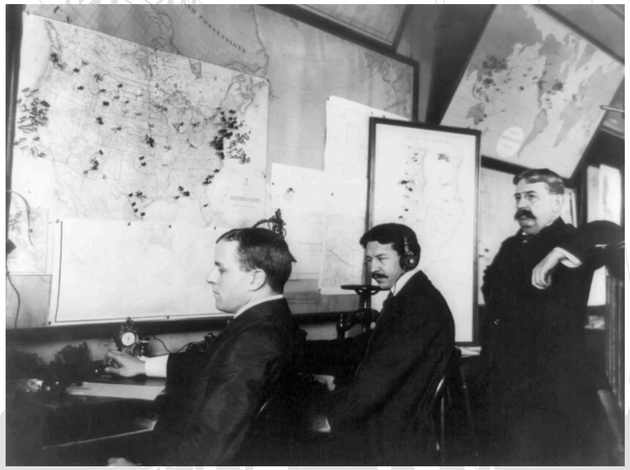
HISTORY OF TRANSMISSION LINE

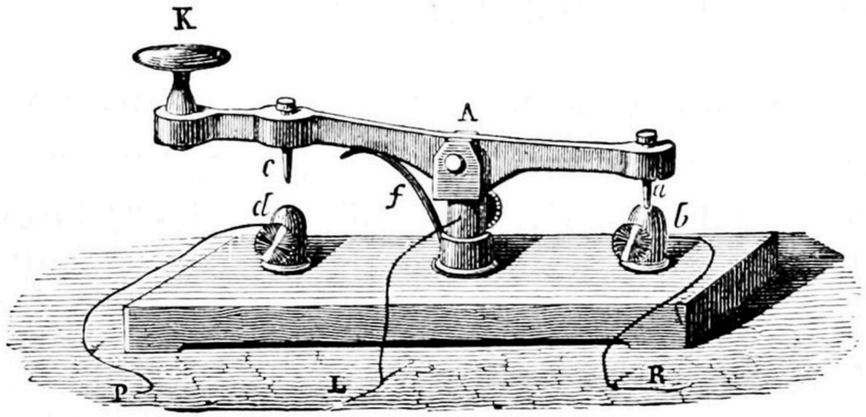
Origin: Telegraph as First Long-Distance Transmission

- Before transmitting electric power, humans transmitted information using electricity.
- 1837 – Samuel Morse invented the electric telegraph.
- First successful line: Washington → Baltimore (1844, ~60 km) with the famous

message:
“What hath God wrought?”

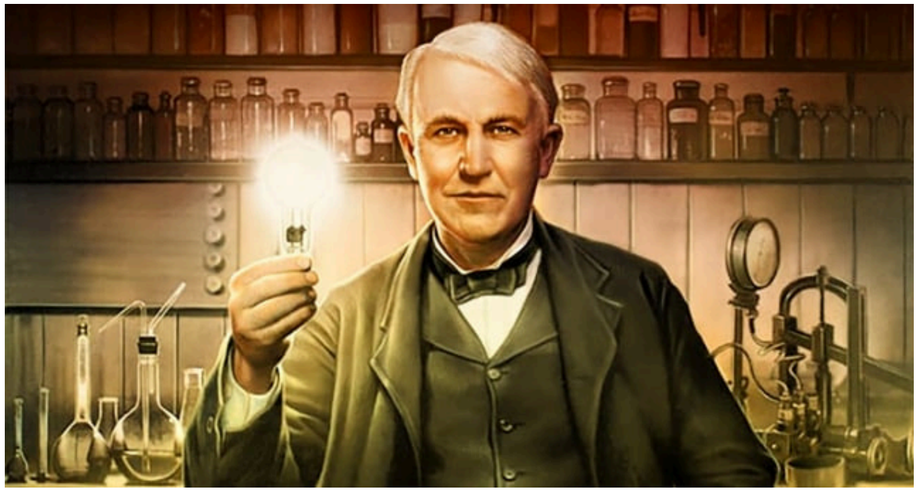


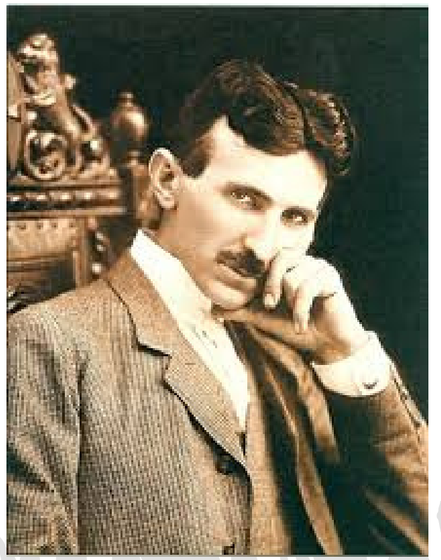


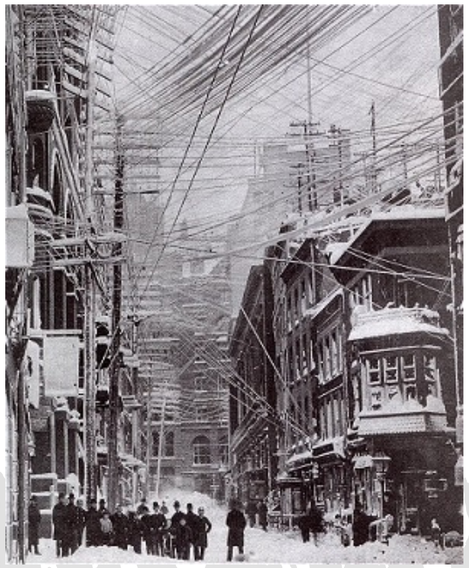


HISTORY OF TRANSMISSION LINE

- 1882: First power station by Thomas Edison in New York (DC system).
- 1886: First long-distance AC transmission line in Oregon (14 miles, 4 kV).
- 1891: Frankfurt Exhibition – 175 km, 25 kV three-phase AC line (Germany).
- 20th century: Rapid expansion of high-voltage AC (HVAC).
- Today: UHV (Ultra High Voltage) AC & HVDC systems (±1100 kV DC in China).







TYPES OF TRANSMISSION LINE

Based on Voltage

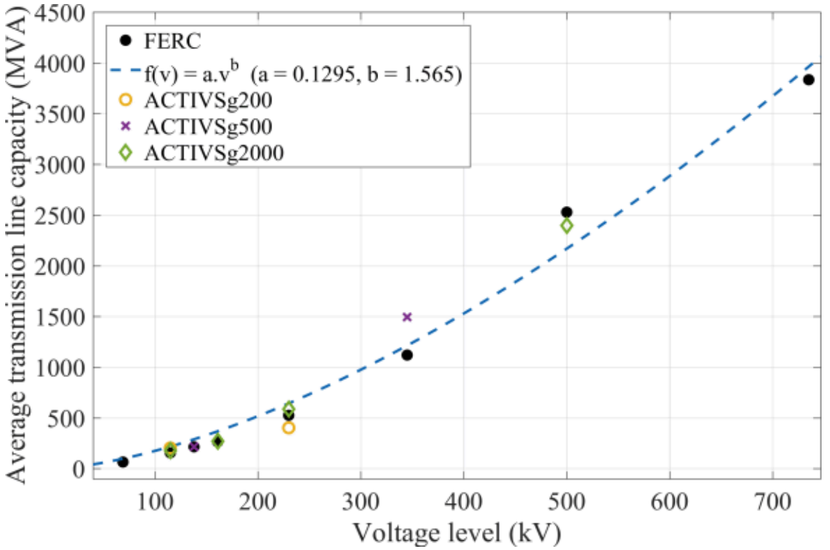
Low Voltage: < 1 kV
High Voltage (HV): 11 – 66 kV
Extra High Voltage (EHV): 132 – 400 kV
Ultra High Voltage (UHV): > 800 kV
Based on Length

Based on Length

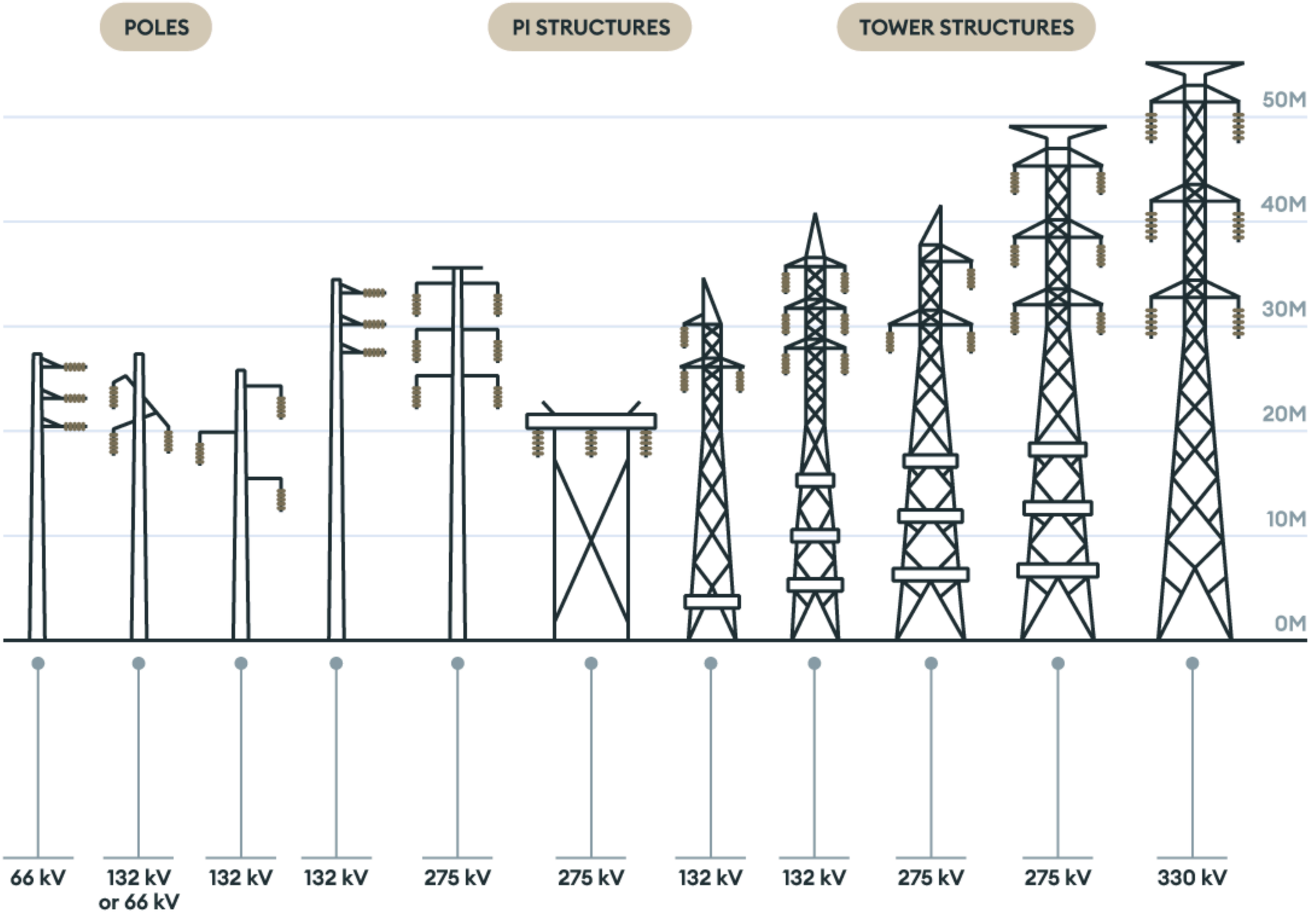
Short line: < 80 km
Medium line: 80 – 250 km
Long line: > 250 km
Based on Current Type

Based on Current Type

AC Transmission Line
DC Transmission Line (HVDC)



Voltage Level	Value Level Mark	System	Valid Section
Low Voltage Level	Below 1000V	AC	Secondary distribution
Medium Voltage Level	1000V to 69 kV	AC	Primary distribution
High Voltage Level	Below 100 kV	AC	Secondary distribution
Extra High Voltage Level	230 kV to 800 kV	AC, DC both	Primary transmission
Ultra High Voltage Level	800 kV to 1000 kV	AC, DC both	Primary transmission
	Over 1000 kV	HVDC is preferable	Primary transmission

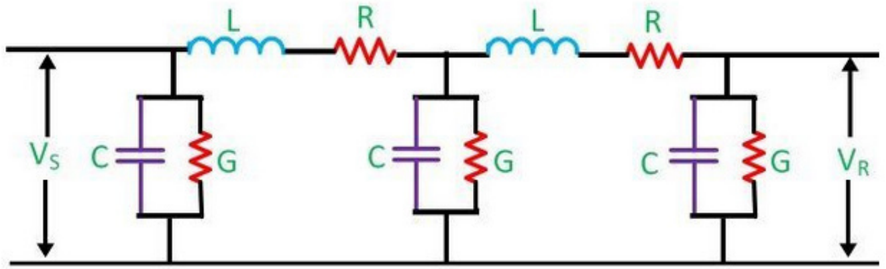

POLES
PI STRUCTURES
TOWER STRUCTURES
50M
40M
30M
20M
10M
0M
66 kV
132 kV or 66 kV
132 kV
132 kV
275 kV
275 kV
132 kV
132 kV
275 kV
275 kV
330 kV


@newt_sch
abbey jaldi bol kal subah
panvel nikalna hai

FACTOR AFFECTING THE TRANSMISSION LINE

Resistance (R):

- Due to conductor material & length.
- Causes I^2R losses (heating).



Inductance (L):

- Due to flux linkage around conductors.
- Leads to reactive power, voltage drops.

$$R = \frac{\rho L}{A}$$

ρ = resistivity
L = length
A = cross sectional area

Capacitance (C):

- Between conductors & ground.
- Causes charging currents, Ferranti effect (voltage rise at receiving end).

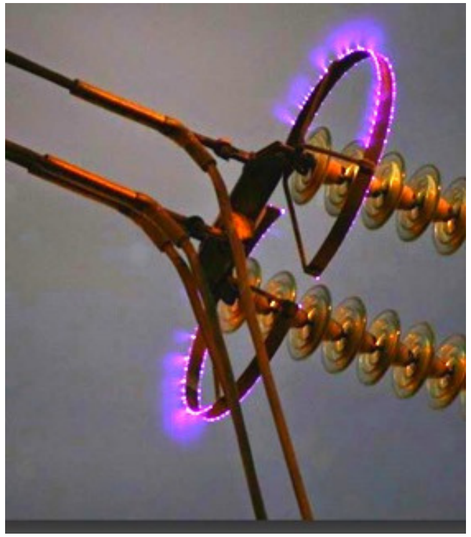
$$= 4 \times 10^{-7} In \frac{D}{r'} \; H/m$$

$$C_{ab} = \frac{q_a}{V_{ab}} = \frac{\pi\varepsilon}{In\frac{D}{r}} F/m$$

FACTOR AFFECTING THE TRANSMISSION LINE

Corona Effect:

- Ionization of air around conductors at high voltage.
- Causes power loss, noise, radio interference, purple glow.
- Controlled by using smooth (hollow/bundled) conductors.

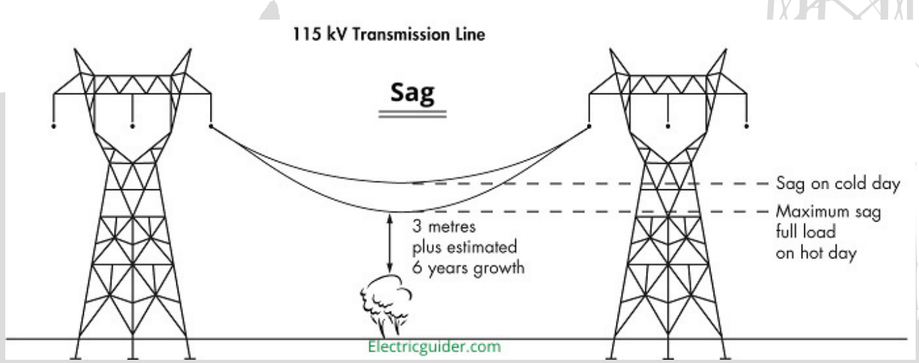


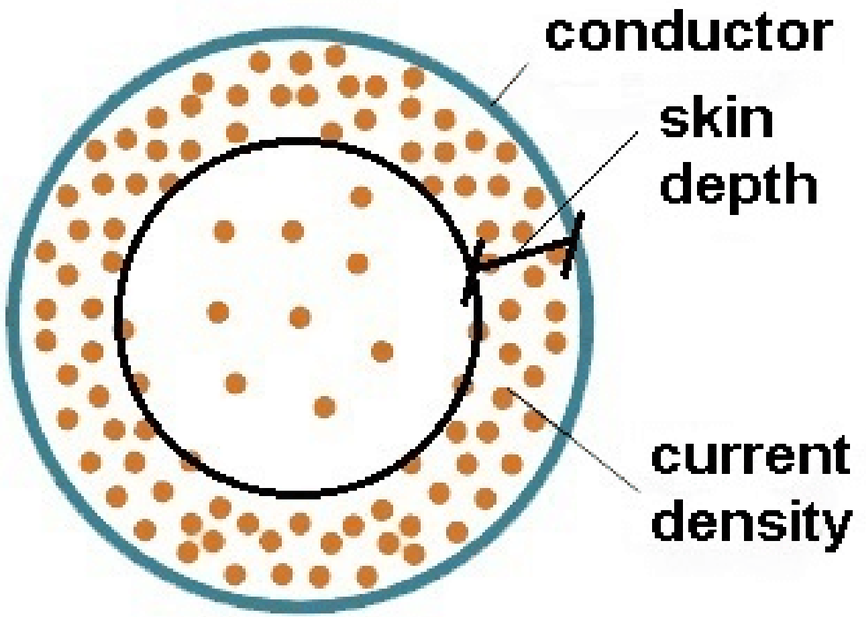
Skin Effect:

- At high frequency, current flows on surface → effective resistance increases.
- More severe in AC (not in DC).

Sag & Tension:

- Conductors expand with heat, so a certain sag is allowed.
- If too high → mechanical stress.
- If too low → clearance problems with ground/trees.





HOW TO REDUCES THE POWER LOSSES

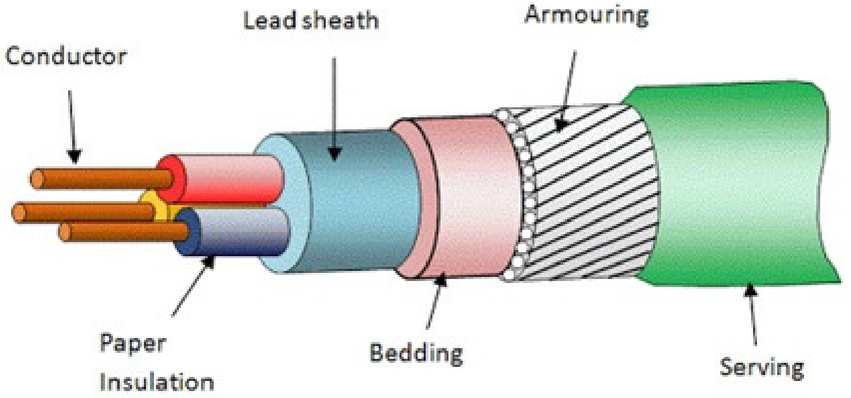
Ohm's Law

$$I = \frac{V}{R}$$

Electric current = Voltage / Resistance



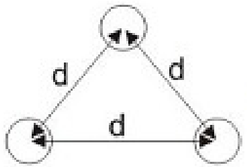




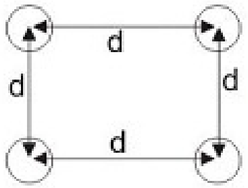




$$GMR = \sqrt[4]{(0.7788r)(d)(d)(0.7788r)} = 0.8825\sqrt{rd}$$



$$GMR = \sqrt[9]{(0.7788r \times d \times d)^3} = 0.92\sqrt[3]{rd^2}$$



$$GMR = \sqrt[16]{(0.7788r \times d \times d \times \sqrt{2}d)^4} = 1.02\sqrt[4]{rd^3}$$



WHAT HAPPEN IF YOU DO NOT TAKE THE GIRD SERIOUSLY



2004 BLACKOUT USA





THANKYOU

EVERYONE

MAY BE THAT IS NOT BORRING

AUDIANCE BE LIKE

